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GT-Finder: Classify the family of glucose transporters with pre-trained BERT language models

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ABSTRACT

Recently, language representation models have drawn a lot of attention in the field of natural language processing (NLP) due to their remarkable results. Among them, BERT (Bidirectional Encoder Representations from Transformers) has proven to be a simple, yet powerful language model that has achieved novel state-of-the-art performance. BERT adopted the concept of contextualized word embeddings to capture the semantics and context in which words appear. We utilized pre-trained BERT models to extract features from protein sequences for discriminating three families of glucose transporters: the major facilitator superfamily of glucose transporters (GLUTs), the sodium-glucose linked transporters (SGLTs), and the sugars will eventually be exported transporters (SWEETs). We treated protein sequences as sentences and transformed them into fixed-length meaningful vectors where a 768- or 1024-dimensional vector represents each amino acid. We observed that BERT-Base and BERT-Large models improved the performance by more than 4% in terms of average sensitivity and Matthews correlation coefficient (MCC), indicating the efficiency of this approach. We also developed a bidirectional transformer-based protein model (TransportersBERT) for comparison with existing pre-trained BERT models.

1. Introduction

Glucose is a vital source of energy for all mammalian cells [1]. It is the primary fuel that provides energy to carry out several human activities, and carbohydrates are the main source of glucose in the human diet [2]. Because of its large size and polar nature [3,4], glucose is impermeable to the hydrophobic cell membrane [3,5]. Therefore, membrane proteins known as glucose transporters mediate the cellular uptake of this essential nutrient across the cell membrane [5,6]. As illustrated in Fig. 1, glucose transporters accomplish the transport of glucose through the plasma membrane into the intracellular fluid. So far, three families of glucose transporters have been identified in humans: the major facilitator superfamily of glucose transporters (GLUTs), the sodium-glucose linked transporters (SGLTs), and the sugars will eventually be exported transporters (SWEETs) [7,8].

GLUT transporters facilitate the absorption of glucose across the lipid bilayer. These transporters are important targets for cancer therapy. In particular, GLUT1, GLUT3, and GLUT5 are essential players in reducing the development of cancer [9]. Dysregulation of GLUT may be linked with many harmful diseases, such as De Vivo disease (GLUT1 deficiency syndrome), Alzheimer's disease, Fanconi-Bickel syndrome, and type 2 diabetes mellitus [10].

SGLT transporters are involved in many vital physiological functions in the human body and act as urea channels, water channels, urea transporters, and glucose sensors [11]. These transporters are also involved in various types of diseases, including cardiac disease, intestinal ischemia, cerebral ischemia, and cancer. The SGLT family of glucose transporters is gaining attention as an attractive therapeutic target in treating various diseases [12].

SWEETs are integral membrane proteins involved in sugar export [8]. These proteins are found in various organisms, including humans, plants, fungi, and bacteria. They are important for metabolism, growth, and homeostasis [13]. SWEETs mainly appear in plants as a new family of glucose transporters [7,8]. The human genome contains only a single member, SWEET1 (or SLC50A1), of the SWEET family of transporters [8].

Glucose transporters perform diverse biological roles. These transport proteins are essential for various functions in the brain [14]. The brain relies on a continuous uptake of glucose and consumes approximately 125 g of glucose per day [15]. The huge demand for energy by

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neurons in the brain is fulfilled by D-glucose with the supply of blood that is facilitated by glucose transporters in the capillaries and brain cells [14]. Glucose transporters also assist the tubular system of the kidney to reabsorb the significant quantity of glucose filtered by the kidneys [15].

It has been shown that language model pre-training is an efficient and effective approach to upgrade the performance of several tasks in the field of natural language processing (NLP). These are sentence-level tasks like natural language inference [16,17] and paraphrasing [18] that attempt to predict the association between various sentences and token-level tasks like question answering [19] and named entity recognition [20]. Pre-training language representations can be implemented to subsequent similar tasks using feature-oriented and fine-tuning techniques. Feature-oriented techniques like ELMo [21] implement architectures that are related to particular tasks by utilizing pre-training representations as additional features. Fine-tuning techniques, such as OpenAI GPT [22], incorporate the idea of task-dependent minimal parameters and are applied by fine-tuning all parameters to the downstream tasks. These two techniques are based on language models that operate in a unidirectional manner for training language representations.

Google developed a state-of-the-art contextualized word embedding technique known as Bidirectional Encoder Representations from Transformers (BERT) [23] to generate deeply bidirectional representations. BERT has proven to be a simple, yet powerful language model that achieved advanced performance on eleven NLP tasks. It achieved a score of 80.5% on the General Language Understanding Evaluation (GLUE), an increase in the score of 7.7%. BERT obtained an accuracy of 86.7% for MultiNLI, improving the accuracy by 4.6%. BERT also achieved scores of 93.2 and 83.1 on SQuAD versions 1.1 and 2.0, raising the scores by 1.5 and 5.1, respectively [23]. Shortly after its publication, the paper [23] obtained more than 10,000 citations from different sources.

BERT is the first-ever model trained on masked language modeling (MLM) in which some input tokens are randomly masked, and the model is trained to predict the masked tokens based on the original context in which they appear in sentences. Compared to pre-trained unidirectional models that only consider left-to-right perspective, MLM enables left and right contexts simultaneously to pre-train a transformer that is completely bidirectional [23].

In this study, we proposed GT-Finder that uses meaningful features generated from various pre-trained BERT models with a support vector machine (SVM) algorithm for discriminating three families of glucose transporters. Our proposed method shows accuracies of 92.43%, 97.30%, and 92.97% for the GLUT, SGLT, and SWEET families of glucose transporters, respectively.

We summarize our contributions as follows: First, we are the first group to use a machine learning technique with enhanced performance to classify glucose transporters into three families. Second, we are also the first group to use a contextualized word embedding technique for this type of transporter family classification. Third, our study assists researchers in bioinformatics in applying NLP-based language models to generate meaningful features for protein classification problems.

2. Related works

There exist a number of computational studies on transporters. Gromiha and Yabuki [24] employed various machine learning techniques to categorize classes of transport proteins by analyzing characteristic features of amino acids. Li et al. [25] proposed a nearest neighbor approach and developed a machine learning framework by integrating homologous search and topological analysis to discriminate transport protein families available in the Transporter Classification Database [26]. Ou et al. [27] utilized the radial basis function network method with position-specific scoring matrices to discriminate three classes and six families of transport proteins. Regarding specific studies on glucose transporters, Burggraaff, et al. [28] proposed random forest–based models to identify SGLT1 inhibitors in silico. In their study, they employed proteochemometrics by incorporating information at the compound and protein levels into random forest models. Nguyen et al. [29] employed SVM with word embeddings to classify transporters into seven substrate specificities, including sugar transporters.

One promising and well-studied method to overcome biological sequence classification problems is to use word or token embedding vectors from language models. For instance, in Refs. [29,30], Nguyen et al. used word embedding-based vectors with promising performance as features to predict transporter specificities and tumor necrosis factors. The word embedding technique can also be seen in studies on classifying shorter sequences, such as enhancer sequences [31] and antimicrobial peptides [32]. It is noteworthy that these studies only employed static word embeddings, which do not take into account all contexts of the sequences. Compared to static word embeddings generated by well-known tools, such as FastText [33] and Word2Vec [34], context-dependent embedding vectors for the same word are not the same in different contexts. We hypothesized that this kind of word embedding would contain more meaningful information for protein sequence classification tasks than static word embeddings. Therefore, we adopted context-dependent embedding vectors for the glucose transporter classification problem. First, we used BERT, an open-source NLP pre-trained language model from Google [23] based on the transformer [35] architecture to generate contextual word embeddings. This is appropriate because the BERT models were trained on natural language to pack the semantic and syntactic information into the contextual word embeddings. These word embeddings were then employed as input features for an advanced classification algorithm, namely, a support vector machine (SVM). We tried various pre-trained BERT models to compare the efficiency of each. In addition, to determine the

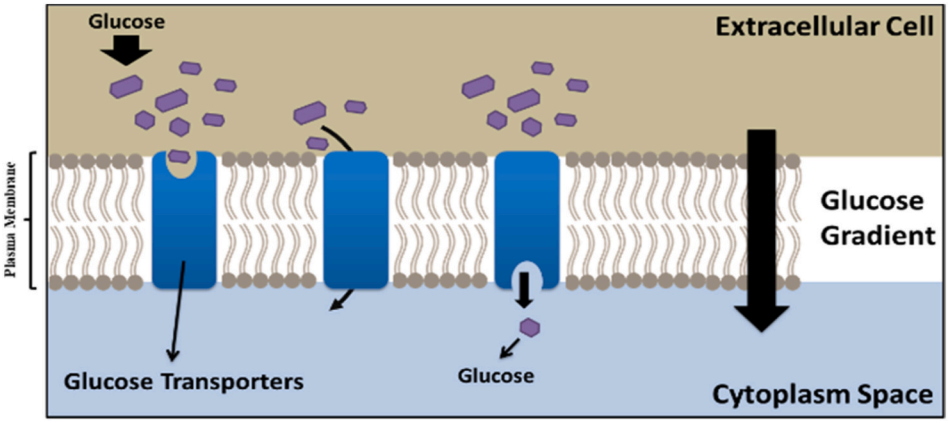


Fig. 1. Representation of glucose transporters in the cell membrane.

effectiveness of BERT models trained on protein language, we also constructed TransportersBERT, a bidirectional, transformer-based protein language model for comparison with pre-trained BERT models that are already available.

3. Method

BERT is an entirely bidirectional unsupervised language representation model that has obtained outstanding performance on eleven NLP tasks. Because of the huge success of BERT, various studies have adopted this technique for their particular tasks. Alsentzer et al. [36] utilized the BERT-Base to generate contextual word embedding models for clinical text. Su et al. [37] introduced representations that were particularly pre-trained for visual-linguistic tasks by implementing two models based on BERT-Base and BERT-Large models. Yung Li [38] proposed a method for extractive summarization by implementing a BERT-Base (Uncased) model. Liu and Lapata [39] introduced a new document level encoder that adopted a BERT-Base (Uncased) model for expressing the semantics of a document to produce sentence-level representations. Shi and Lin [40] used BERT-Base (Cased) and BERT-Large (Cased) models for relation extraction and semantic role labeling. Adhikari et al. [41] adopted BERT-Base and BERT-Large models for document classification tasks. Levine et al. [42] proposed a method for applying self-supervision directly on the word sense level. They introduced two models derived from the original BERT-Base and BERT-Large models. Joshi et al. [43] introduced a pre-trained method that uses configurational settings of BERT-Large to predict the span of text.

Fig. 2 illustrates the workflow of our study, including the data gathering, preprocessing, classification, and evaluation phases. The details of each phase are described below.

3.1. Dataset collection

We collected data for three families of glucose transporters, namely, GLUT, SGLT, and SWEET, from the protein database of the National Center for Biotechnology Information (NCBI) [44], which links to numerous data sources. The sources include the Protein Data Bank (PDB) [45], the Reference Sequence (RefSeq) database [46], the UniProtKB/Swiss-Prot database [47] (only manually annotated and reviewed protein sequences were considered), the DNA Data Bank of Japan (DDBJ) [48], the European Molecular Biology Laboratory (EMBL) database [49] and the Gen Bank (GenBank) sequence database [50]. All of these data sources have a significant number of protein sequences for the three families of glucose transporters. The PDB database contains 63, 6, and 15 protein sequences for GLUT, SGLT, and SWEET, respectively. RefSeq has 11303 GLUT, 58 SGLT, and 1220 SWEET protein sequences. UniProtKB (Swiss-Prot) has 152, 20, and 66 proteins for GLUT, SGLT, and SWEET, respectively. DDBJ consists of 98 GLUT, 176 SGLT, and 43 SWEET sequences. EMBL contains 227 proteins for GLUT, 404 proteins for SGLT, and 133 proteins for SWEET. GenBank consists of 3211, 3487, and 991 sequences for the three families. Therefore, the overall dataset contains 15054 sequences for GLUT, 4151 sequences for SGLT, and 2468 sequences for the SWEET family of glucose transporters. We found a number of identical protein sequences in the three families of glucose transporters because the data were gathered from various heterogeneous sources; therefore, we removed the redundant protein sequences. Similarly, we only considered protein sequences having a length greater than 50 for the three families of glucose transport proteins. Additionally, protein sequences from three families of glucose transporters also contain ambiguous terms, such as *like*, *similar*, *partial*, *isoform*, *predicted*, and *precursor*. Those protein sequences were also exempted from the three datasets for further processing. After performing the preliminary preprocessing procedure, we were left with 15749 protein sequences in total. Among them, GLUT, SGLT, and SWEET glucose transporter families contain 9616, 4107, and 2026 proteins, respectively. We next applied the CD-HIT program [51] to remove protein sequences with more than 40% identity. The resulting datasets have 510, 225, and 190 proteins for the GLUT, SGLT, and SWEET families, respectively. The dataset of each family was randomly partitioned into testing and training portions in a 20:80 ratio. The training portion was used to build the model and the testing portion to evaluate its performance. Table 1 displays the statistics of the surveyed data.

3.2. Feature extraction using contextualized word embeddings from pre-trained BERT models

Pre-trained language models, such as ELMo (Embeddings from Language Models) [21] and BERT [23], have achieved remarkable improvements in numerous NLP tasks and have outperformed other language models, including natural language inferencing, question answering, machine translation, etc. Pre-trained BERT models can be used for different NLP tasks such as feature generation. The features created from BERT can be utilized and fine-tuned for the NLP tasks of interest, such as text classification, next-sentence prediction, document summarization, and more. In this process, previously available knowledge from a large text corpus can be transferred to accomplish tasks with less abundant data or in other domains. Thus, we attempted to exploit both advantages that pre-trained BERT models offer: the huge amount of information for training BERT and the domain switch from natural to protein language. Specifically, we transformed each protein sequence into a "protein sentence" composed of uni-grams (a single amino acid) to be input into the pre-trained BERT models. We imposed a maximum length of 510 on the sequences because all BERT models accept sequences with a maximum length of 512 and add two special tokens (a CLS token and a SEP token) at the beginning and end of each sequence, respectively. Therefore, we only considered the first 510 amino acids, at most, from a long protein sequence, while the shorter protein sequences were used with their original lengths.

As mentioned in the original BERT paper [23], BooksCorpus, containing 800 million words, and English Wikipedia, containing 2500 million words, were used as the pre-training corpus for all variants of BERT. We used these pre-trained models to generate important context-based representations for the three families of glucose transport proteins. BERT [23] was released with various pre-trained models with specific configurations. BERT-Base cased and uncased models contain 12 layers (encoders or transformer blocks), 768 hidden units, 12 heads, and 110 million parameters. BERT-Large cased and uncased both have 24 layers (encoders or transformer blocks), 1024 hidden units, 16 heads, and 340 million parameters. In this study, we only carried out experiments with BERT-Base (cased and uncased) and BERT-Large (cased and uncased) models. Uni-gram protein sequences with a maximum length of 510 were input into the BERT models, and feature vectors for each of the 12 layers were generated for each amino acid residue. The feature vectors have a fixed length of 768 for the BERT-Base and 1024 for the BERT-Large pre-trained models. During the experiments, some of the layers were combined in various arrangements to obtain the most suitable representations. The arrangements included concatenating the last four layers, summing up the last four layers, summing up all 12 layers, or just selecting the last hidden encoder layer. In the present study, we adopted the last hidden encoder layer generated by BERT as a feature representation vector for each amino acid of the surveyed protein sequences. Next, we summed up the feature vectors corresponding to the same amino acid in a protein sequence, and each element of the resultant vector was divided by the length of the sequence. Finally, we created the final feature vectors by appending the vector of each amino acid in a specific and consistent order. Thus, the final feature vectors have 20×768 or 20×1024 elements. Fig. 3 illustrates the complete process of feature generation for the BERT models. For more details about the mechanism of BERT, refer to Fig. S1, Section A in the supporting information.

Transfer learning refers to the transformation of knowledge from one domain or task that has already been learned to other similar domains or

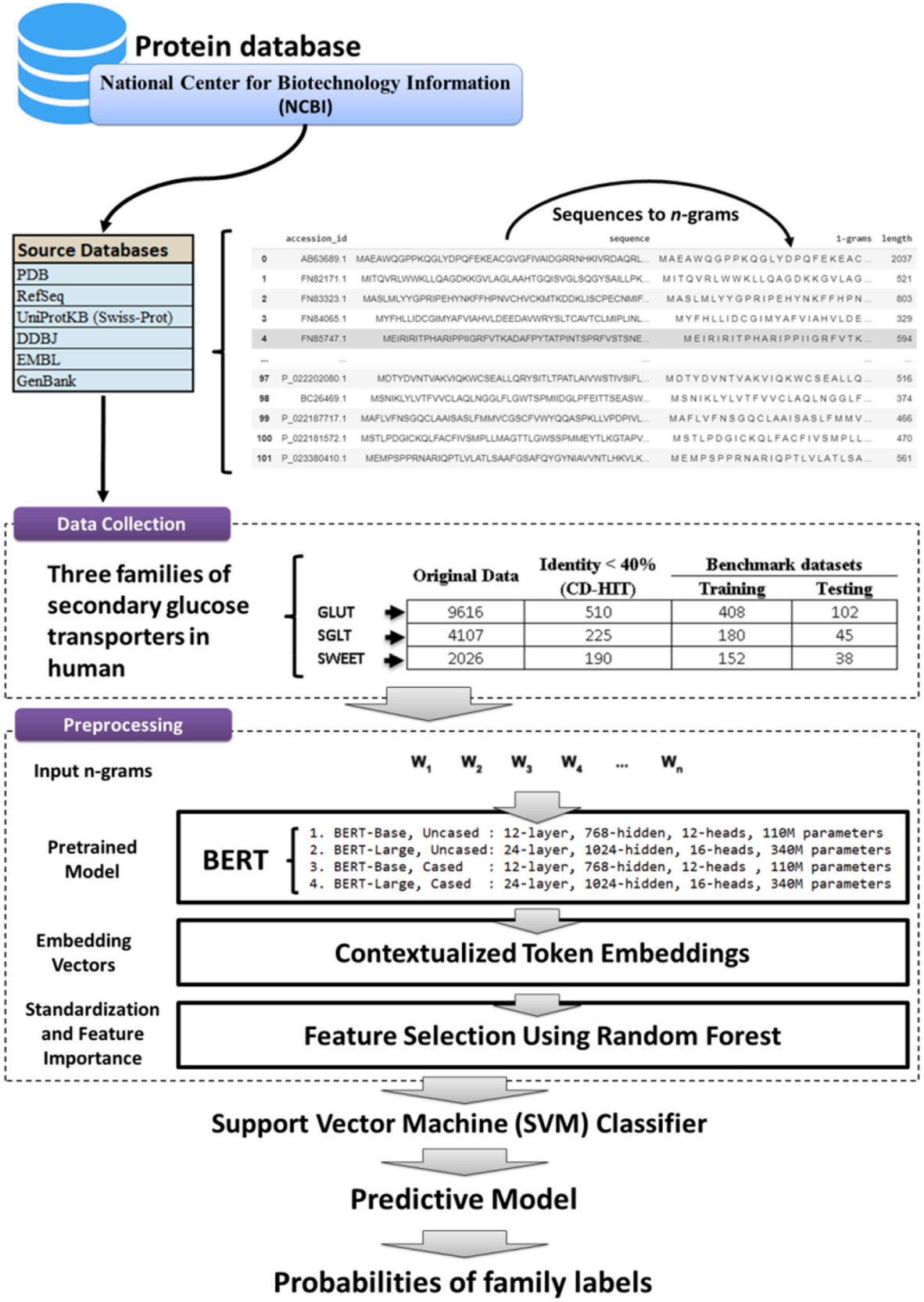


Fig. 2. The workflow of our study, including the data gathering, preprocessing, classification, and evaluation phases.

Table 1
Summary of benchmark datasets including testing data and training data.

Datasets	Original Data	Similarity <40% (CD-HIT)	Benchmark datasets	
			Testing Data	Training Data
GLUT	9616	510	102	408
SGLT	4107	225	45	180
SWEET	2026	190	38	152
Total	**15749**	**925**	**185**	**740**

tasks to improve learning capability [52]. Transfer learning has become popular and has been extensively adopted in machine learning and deep learning applications, and it has a variety of applications in other fields [53]. In the present study, we applied transfer learning by transferring the knowledge learned by BERT from the English language to the protein language for extracting meaningful features from protein sequences.

3.3. Feature normalization and standardization

To accelerate the calculations of algorithms used in classification tasks, we applied a normalization and standardization step to the feature vectors created from previous steps. We used StandardScaler() [54], which is one of the possible scalers available for feature scaling (standardization) of data. It normalized the features in the dataset so that the mean is equal to zero and the standard deviation is equal to one. It is calculated as

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where z is the final transformed value, x is the original value, μ is the mean, and σ is the standard deviation.

3.4. Dataset balancing

To classify three families of glucose transporters (GLUT, SGLT, and SWEET), we applied a binary classification technique to assemble the data into positive and negative instances. In this way, our datasets are contained in three binary classes. For each binary class, one family of glucose transporters is labeled as a positive dataset while the other families are labeled as negative datasets. Thus, the number of positive instances in each problem is much less than the number of negative instances. In other words, our datasets are imbalanced. Therefore, we

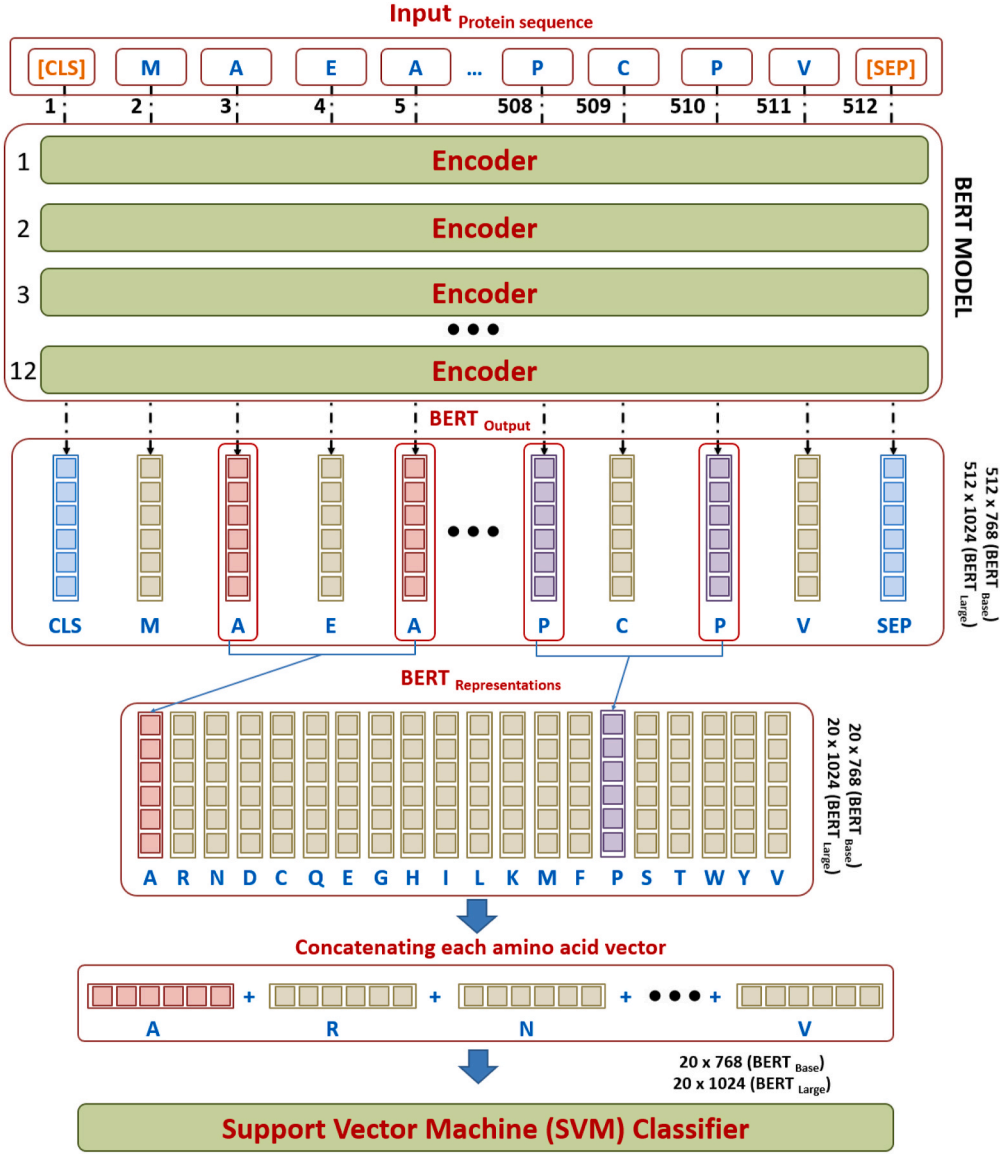


Fig. 3. The complete process for generating features using pre-trained BERT models.

employed the syntactic minority oversampling technique (SMOTE) [55] to resolve the issue of imbalanced datasets.

3.5. Feature selection with random forests

The objective of feature selection is to select a smaller subset of features that contains most of the relevant information that best represents the whole feature set [56]. We adopted random forests [57] as a feature selection technique for identifying meaningful features from the complete dataset.

Random forests compute feature importance using Gini importance [58] or mean decrease in impurity. It is possible to calculate the degree to which the impurity is reduced for each feature when training a tree. This indicates how much a model fit or accuracy decreases when a feature is removed from the analysis. A particular feature is considered important if it decreases the impurity to its minimum level. The mean decrease is a useful parameter for selecting important features. In a random forest, the ultimate importance of a feature is determined by taking the average decrease in impurity across all trees in the forest.

3.6. Classification with support vector machine

We used a support vector machine (SVM) classifier to perform the classification. The SVM has been used as the primary classifier in many bioinformatics problems due to its efficiency. There are many kernel function options in SVM, and we chose the radial basis function kernel to better classify high dimensional data. We also searched for the optimal hyperparameters (cost and gamma) before any experiments.

3.7. Environmental settings

The experiments were performed on Google Colaboratory (Google Colab) [59] with Python 3 in the background. Colab is a free notebook offered by Google, similar to the Jupyter notebook. It provides up to Tesla K80 GPU for free to execute code segments in the Cloud.

3.8. Feature extraction from contextualized word embedding of TransportersBERT

In this work, we have also created TransportersBERT, a BERT model trained on protein sequence data available at Transporters Classification Database (TCDB) and freely accessible via *http://tcdb.org*. At the time of our experiments, 18,880 protein sequences existed in the database. We constructed TransportersBERT considering the same configuration as the BERT-Base (Uncased) model. The goal of this experiment is to compare the effectiveness of context-dependent word embedding vectors from different pre-trained BERT models. The uni-gram sentences, having a maximum length of 510 as described in the previous section, went through the same procedures with TransportersBERT to generate final feature vectors.

3.9. Evaluation metrics for classification

We utilized sensitivity (Sens), specificity (Spec), overall accuracy (Overall Acc), balanced accuracy (Bal Acc), and Matthews correlation coefficient (MCC) as the measurement metrics to evaluate model performance. These metrics were calculated using the following equations.

$$Sens = \frac{TP}{TP + FN} \tag{2}$$

$$Spec = \frac{TN}{TN + FP} \tag{3}$$

$$Overall\ Acc = \frac{TP + TN}{TP + FP + TN + FN} \tag{4}$$

$$Bal\ Acc = \left(\frac{TP}{TP + FN} + \frac{TN}{TN + FP}\right) \Big/ 2 \tag{5}$$

$$MCC = \frac{(TP\ x\ TN) - (FP\ x\ FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{6}$$

where TP represents the number of true positives, FN the number of false negatives, TN the number of true negatives, and FP the number of false positives. Sensitivity measures the proportion of actual positive instances correctly identified as the class of interest; therefore, it is the true positive rate (TPR), while specificity is the true negative rate (TNR) representing the proportion of negative data items recognized accurately.

3.10. Analysis using receiver operating characteristics (ROC) curve

We employed the receiver operating characteristics (ROC) curve and the area under the curve (AUC) as our statistical measurements. ROC is a widely accepted machine learning and data mining technique. It is a graphical representation that shows the relationship between the true-positive rate (sensitivity) and false-negative rate (1-specificity). It illustrates the overall efficiency of a classification model for all potentially correct and incorrect decisions at varying decision thresholds [60,61]. Compared to the ordinary performance evaluation indices, ROC provides a visual and numerical description of the behavior of a classifier. Additionally, in the bioinformatics domain, it is becoming a prevalent technique for assessing the significance of various classifiers [62]. Similarly, the area under the curve, most often referred to as AUC, has received much attention in the machine learning community and is the most popular performance measure in the supervised learning domain. It is widely adopted by numerous studies as an analysis tool for visualizing, comparing, and assessing the significance of various predictors. The AUC is an individual score in the interval from 0 to 1 and can be described as the probability based on whichever randomly selected positive instance can be assigned a higher rank than a randomly selected negative instance. Therefore, the performance evaluation of a classifier with a larger AUC score is considered as favorable [63].

Fig. 4 illustrates the AUC and ROC curve as a statistical analysis to characterize the predictive performance of the GLUT, SGLT, and SWEET glucose transporters using BERT-Large (Uncased), BERT-Base (Uncased), and BERT-Large (Uncased), respectively. Fig. 4 indicates that the SVM classifier obtained higher AUC scores of 0.93, 0.94, and 0.90 for GLUT, SGLT, and SWEET, respectively, compared to other traditional machine learning algorithms. Furthermore, Fig. 4 shows that the SVM classifier yields a larger area under the ROC curves for all three families of glucose transporters.

4. Results

In this section, first, we compared the performance of the various proposed BERT models for three families of glucose transporters. Next, we evaluated the performance of several traditional machine learning algorithms on classifying three families of glucose transporters using the best BERT features. Finally, we compared the outcomes of some conventional feature extraction methods, including amino acid composition (AAC) [27,64–67] and dipeptide composition (DPC) [27,64,66,67] with our newly proposed feature extraction method.

4.1. Performance comparison of contextualized word embedding features from various pre-trained BERT models

In these experiments, we compared the effectiveness of contextual word embedding features from different pre-trained BERT models, such as TransportersBERT (Uncased), BERT-Base (Cased, Uncased), and BERT-Large (Cased, Uncased) for classifying three families of glucose

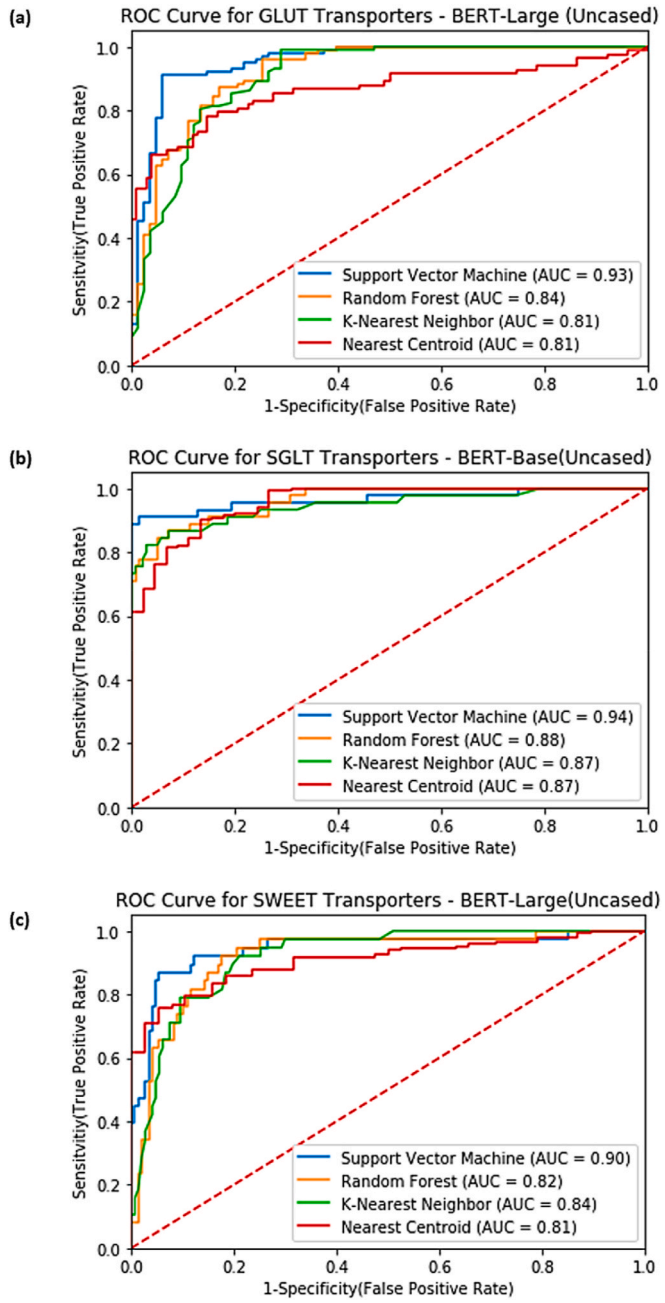


Fig. 4. Receiver operating characteristic (ROC) curves for demonstrating the significance of the proposed method with various machine learning classifiers: (a) GLUT Transporters, (b) SGLT Transporters, and (c) SWEET Transporters.

transporters. Researchers often use a five-fold or ten-fold cross-validation technique for evaluating the effectiveness of different types of features on training data and then assess the performance on independent test data. However, considering the large amount of data for each binary classification problem, we only used the training data for generating different features from various pre-trained BERT models, as discussed above. After that, we evaluated the independent test performance, the results of which are given in Table 2. Table 2 shows that BERT-Large (Uncased) outperforms all other pre-trained models on the GLUT transporter family by achieving 91.18% for Sens, 93.98% for Spec, 92.43% for Overall Acc, 92.58% for Bal Acc, and 0.85 for MCC. For the SGLT transporters, BERT-Base (Uncased) performs better than the other pre-trained BERT models. It obtains 88.89% for Sens, 100.00% for Spec, 97.30% for Overall Acc, 94.44% for Bal Acc, and 0.93 for MCC.

Table 2
Performance of existing pre-trained BERT models for three families of glucose transporters using the independent dataset with the SVM classifier. (Scores for the best model for each family are marked in bold).

BERT Model	Family	Sens (%)	Spec (%)	Overall Acc (%)	Bal Acc (%)	MCC
TransportersBERT (Uncased)	GLUT	80.39	86.75	83.24	83.57	0.67
BERT-Base (Uncased)		92.16	89.16	90.81	90.66	0.81
BERT-Large (Uncased)		**91.18**	**93.98**	**92.43**	**92.58**	**0.85**
BERT-Base (Cased)		91.18	84.34	88.11	87.76	0.76
BERT-Large (Cased)		90.20	85.54	88.11	87.87	0.76
TransportersBERT (Uncased)	SGLT	88.89	95.71	94.05	92.30	0.84
BERT-Base (Uncased)		**88.89**	**100.00**	**97.30**	**94.44**	**0.93**
BERT-Large (Uncased)		84.44	99.29	95.68	91.87	0.88
BERT-Base (Cased)		91.11	98.57	96.76	94.84	0.91
BERT-Large (Cased)		82.22	97.86	94.05	90.04	0.83
TransportersBERT (Uncased)	SWEET	73.68	85.03	82.70	79.36	0.53
BERT-Base (Uncased)		81.58	95.24	92.43	88.41	0.77
BERT-Large (Uncased)		**84.21**	**95.24**	**92.97**	**89.72**	**0.79**
BERT-Base (Cased)		78.95	95.24	91.89	87.09	0.75
BERT-Large (Cased)		78.95	93.20	90.27	86.07	0.71

BERT-Large (Uncased) has the highest performance of the BERT models on the SWEET transporters with 84.21% for Sens, 95.24% for Spec, 92.97% for Overall Acc, 89.72% for Bal Acc, and 0.79 for MCC.

It is also noteworthy that features from TransportersBERT are not as efficient as those from other pre-trained BERT models. This may be because of insufficient data used in training TransportersBERT. We compare the amount of information used to train existing pre-trained BERT models and our TransportersBERT model in Section B of the supporting information. Furthermore, we used the best feature types: BERT-Large (Uncased) features for GLUT and SWEET, and BERT-Base (Uncased) features for SGLT in further experiments.

Fig. 5 displays the performance of five pre-trained BERT models for the three families of glucose transporters in a bar chart. In each bar charts the primary vertical axis represent sensitivity (Sens), specificity (Spec), overall accuracy (Overall Acc) and balanced accuracy (Bal Acc), whereas, the secondary vertical axis represents Matthews correlation coefficient (MCC). Fig. 5 (a) and (c) show that BERT-Large (Uncased) achieved higher performance for the GLUT and SWEET families of glucose transporters for many predictive performance measures compared to other BERT models. Fig. 5 (b) shows that BERT-Base (Uncased) obtained better performance for many evaluation metrics than the other BERT models.

4.2. Performance of traditional machine learning algorithms in discriminating glucose transporters using the best BERT features

After determining the best feature types for each binary classification problem, we used these features as input into four advanced classification algorithms: Random Forest [68], k-Nearest Neighbor [69], SVM [70], and Nearest Centroid [71].

We evaluated five-fold cross-validation and independent tests on three families of glucose transporters. The results, presented in Table 3, clearly show that SVM achieves higher performance for many metrics on the three families of glucose transporters for the five-fold cross-

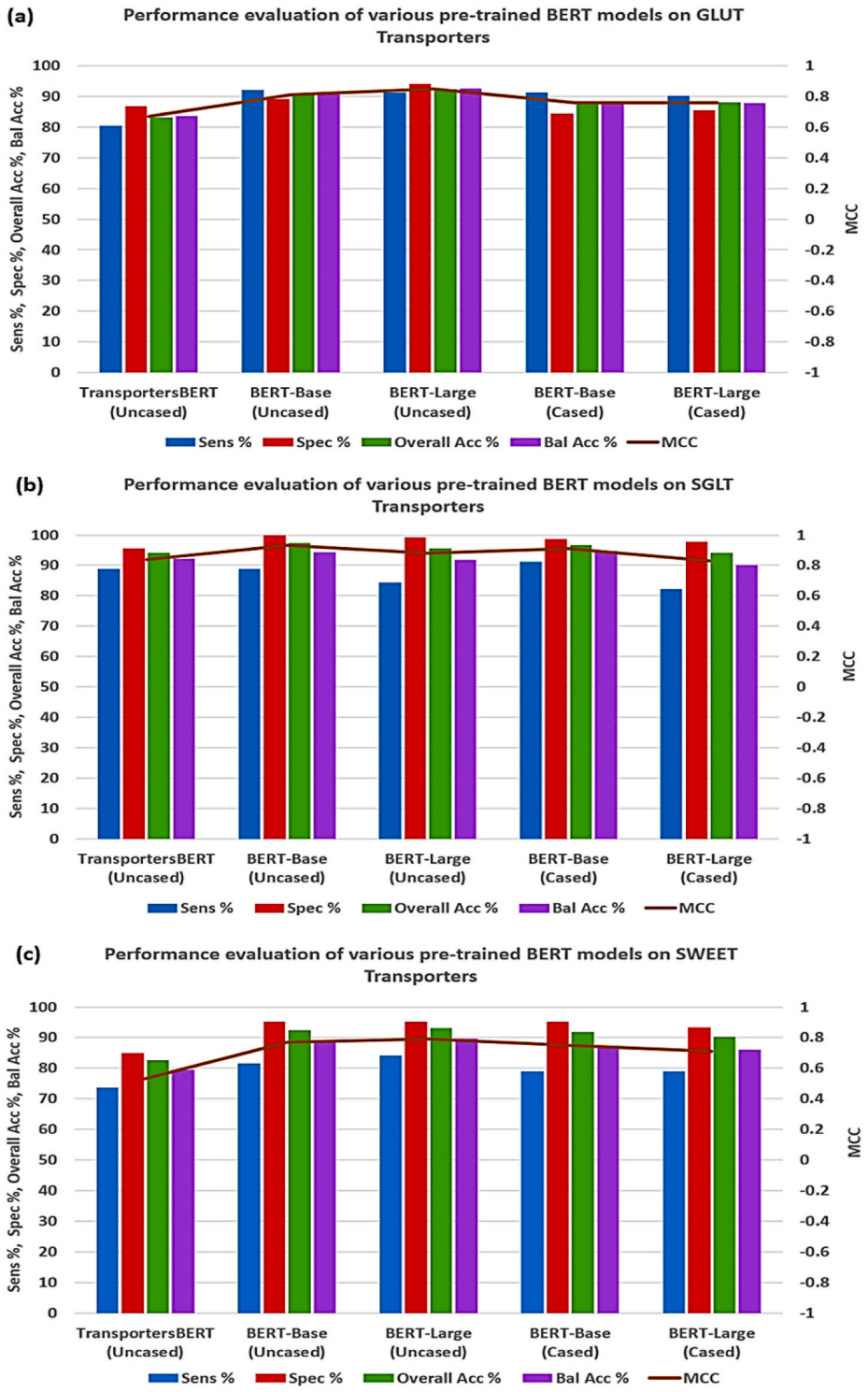


Fig. 5. Performance evaluation of various pre-trained BERT models for the three families of glucose transporters with the independent dataset using the SVM Classifier: (a) GLUT, (b) SGLT, and (c) SWEET.

validation and independent tests compared to other traditional machine learning classifiers. Specifically, on classifying GLUT transporters, SVM obtains a Sens of 87.74%, a Spec of 92.91%, an Overall Acc of 90.32%, a Bal Acc of 90.32%, and an MCC of 0.81. For the SGLT dataset, the SVM outperforms other classifiers and attains 98.75% for Sens, 96.96% for Spec, 97.86% for Overall Acc, 97.86% for Bal Acc, and 0.96 for MCC. For SWEET transporters, SVM obtains 98.57% for Sens, 95.36% for Spec, 96.96% for Overall Acc, 96.96% for Bal Acc, and 0.94 for MCC.

Furthermore, on the independent test datasets, the performance of SVM also surpasses the other classifiers. On the GLUT family of glucose transporters, SVM attains 91.18% for Sens, 93.98% for Spec, 92.43% for Overall Acc, 92.58% for Bal Acc, and 0.85 for MCC. For the SGLT dataset, it achieves scores for Sens, Spec, Overall Acc, Bal Acc, and MCC of 88.89%, 100.00%, 97.30%, 94.44%, and 0.93, respectively. For the SWEET dataset, SVM scored 84.21% for Sens, 95.24% for Spec, 92.97% for Overall Acc, 89.72% for Bal Acc, and 0.79 for MCC. Moreover, it is evident from Table 3 that the SVM model in the SGLT transporters achieves the best performance on both the cross-validation and

Table 3
Performance comparison of 5-fold cross-validation and independent test for various traditional machine learning classifiers on the three data sets. (Scores for the best classifier for each family are marked in bold).

Family	Classifiers	Sens (%)	Spec (%)	Overall Acc (%)	Bal Acc (%)	MCC
5-Fold Cross-Validation						
GLUT	Random Forest	76.70	82.58	79.64	79.64	0.60
	Nearest Neighbors	87.50	69.29	78.40	78.40	0.59
	SVM	**87.74**	**92.91**	**90.32**	**90.32**	**0.81**
	Nearest Centroid	74.75	73.22	73.99	73.99	0.49
SGLT	Random Forest	88.75	86.43	87.59	87.59	0.76
	Nearest Neighbors	91.61	78.21	84.91	84.91	0.71
	SVM	**98.75**	**96.96**	**97.86**	**97.86**	**0.96**
	Nearest Centroid	88.57	83.57	86.07	86.07	0.73
SWEET	Random Forest	80.18	79.46	79.82	79.82	0.60
	Nearest Neighbors	89.82	64.29	77.05	77.05	0.56
	SVM	**98.57**	**95.36**	**96.96**	**96.96**	**0.94**
	Nearest Centroid	85.18	72.50	78.84	78.84	0.59
Independent Test						
GLUT	Random Forest	89.22	78.31	84.32	83.76	0.68
	Nearest Neighbors	89.22	73.49	82.16	81.35	0.64
	SVM	**91.18**	**93.98**	**92.43**	**92.58**	**0.85**
	Nearest Centroid	85.29	77.11	81.62	81.20	0.63
SGLT	Random Forest	77.78	98.57	93.51	88.17	0.82
	Nearest Neighbors	88.89	84.29	85.41	86.59	0.67
	SVM	**88.89**	**100.00**	**97.30**	**94.44**	**0.93**
	Nearest Centroid	82.22	91.43	89.19	86.83	0.72
SWEET	Random Forest	73.68	91.16	87.57	82.42	0.63
	Nearest Neighbors	78.95	88.44	86.49	83.69	0.63
	SVM	**84.21**	**95.24**	**92.97**	**89.72**	**0.79**
	Nearest Centroid	73.68	87.76	84.86	80.72	0.57

independent tests. Therefore, we used the SVM algorithm in further experiments.

4.3. Performance evaluation of our proposed approach and other feature extraction techniques

We employed different feature extraction methods to evaluate the performance of three families of glucose transport proteins. In this work, we selected two well-known feature extraction methods, AAC and DPC, and compared their results with the performance of our newly proposed feature extraction method. Table 4 summarizes the outcomes of the two feature extraction methods and those of our proposed BERT model on independent data. Compared to AAC and DPC features, our proposed BERT models attained superior performance for all the three families of glucose transporters. In particular, for the GLUT family, our SVM model achieves Sens, Spec, Overall Acc, Bal Acc, and MCC values of 91.18%, 93.98%, 92.43%, 92.58%, and 0.85, respectively. Compared to AAC, our model is higher by 8.83%, 9.64%, 9.19%, 9.23%, and 0.19 for Sens, Spec, Overall Acc, Bal Acc, and MCC, respectively. Compared to DPC, the performance of our model is slightly lower for Sens, higher by 5% for Spec, 2% higher for Overall Acc and Bal Acc, and higher by 0.04 for MCC.

Table 4
Performance evaluation of our proposed method and other feature extraction techniques using independent test. (Scores for our proposed method are marked in bold).

Method	Family	Sens (%)	Spec (%)	Overall Acc (%)	Bal Acc (%)	MCC
AAC	GLUT	82.35	84.34	83.24	83.35	0.66
	SGLT	95.56	80.71	84.32	88.13	0.67
	SWEET	68.42	90.48	85.95	79.45	0.58
DPC	GLUT	92.16	89.16	90.81	90.66	0.81
	SGLT	97.78	92.14	93.51	94.96	0.84
	SWEET	52.63	99.32	89.73	75.98	0.66
Proposed Method	GLUT	**91.18**	**93.98**	**92.43**	**92.58**	**0.85**
	SGLT	**88.89**	**100.00**	**97.30**	**94.44**	**0.93**
	SWEET	**84.21**	**95.24**	**92.97**	**89.72**	**0.79**

When compared with the SGLT family, our proposed method achieved a Sens of 88.89%, which is lower than Sens achieved by AAC and DPC, but the Spec for our model is 100%, which is increased by 19.29% and 7.86% for AAC and DPC, respectively. Our proposed model obtained an Overall Acc of 97.30%, improved by 13% and 4% compared to AAC and DPC, respectively, while the MCC score for the proposed model is 0.93, which is higher by 0.26 and 0.09 than AAC and DPC, respectively. Furthermore, the Sens, Spec, Overall Acc, Bal Acc, and MCC of the proposed method on the SWEET family of glucose transporters are 84.21%, 95.24%, 92.97%, 89.72%, 0.79, respectively. Regarding these scores with AAC and DPC, the Sens is improved by 15.79% and 31.58%; Overall Acc is enhanced by 7.02% and 3.24%; Bal Acc is advanced by 10.27% and 13.74%; MCC is raised by 0.21 and 0.13, respectively. It can also be examined that the Spec is increased by 4.76%, and decreased by 4.08% compared with ACC and DPC, respectively.

5. Discrimination on the web

We have constructed a web server available at *http://bio216.bioinfo.yzu.edu.tw/glubert/* for the classification of three families of glucose transporters according to our experiments described above. Using this server, biologists can submit protein sequences belonging to the glucose transporters to examine the corresponding families of these sequences.

6. Conclusions

In this study, we used BERT, a recent language representation model, to generate contextualized word embeddings in protein language for the classification of glucose transporter families. Using pre-trained BERT models, we were able to transfer the syntactic and semantic knowledge from a huge human language text corpus to protein data because of BERT's outstanding performance on different Natural Language Processing tasks as well as the similarities between protein language and human language. Therefore, we transformed our protein data into fixed-length feature vectors. In these representations, 768 and 1024 fixed-length dimensional vectors describe each amino acid residue for BERT-Base and BERT-Large models, respectively. The experimental results show that the combination of features from BERT-Base (uncased) or BERT-Large (uncased) models with the Support Vector Machine classifier generates suitable outcomes for discriminating the three families of glucose transport proteins.

Due to the current input length restriction by BERT, we can only input the first 510 residues from each protein sequence. However, even with this limitation, we have so far achieved significant performance. In future research, we hope to achieve a better solution by using complete sequences as input.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Availability of data and materials

The dataset supporting the conclusions of this article is available at the given link *http://bio216.bioinfo.yzu.edu.tw/glubert/*.

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Authors' contributions

Conceived and designed the study: SMAL, SWT, and YYO. Collected and analyzed the data: SMAL and SWT. Designed and performed the experiments: SMAL. Wrote the paper: SMAL, SWT, and TTDN. Web Server Design: QTH. Manuscript proofreading: TTDN and SMAL. All authors have read and approved this manuscript.

Declaration of competing interest

None.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.compbiomed.2021.104259.

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